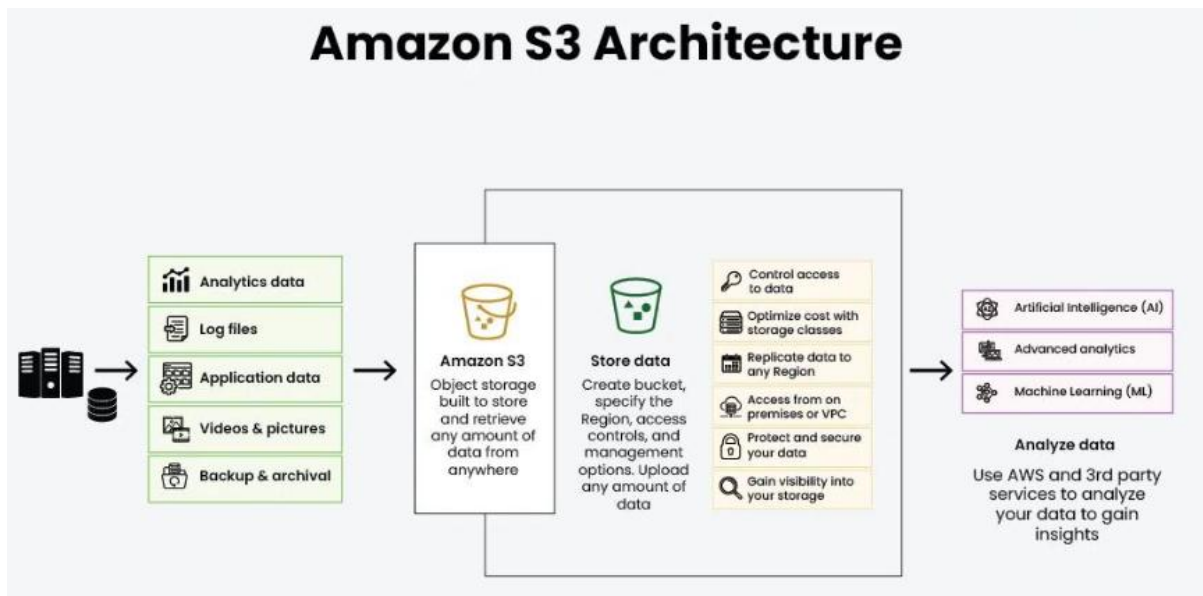


Amazon S3 architecture, buckets, and objects

Amazon Simple Storage Service (S3) is an object storage service that offers industry-leading scalability, data availability, security, and performance. It is the backbone of the AWS ecosystem, used by everyone from tiny startups to massive enterprises like Netflix and Airbnb to store petabytes of data.

Unlike file storage (like EFS) or block storage (like EBS), S3 stores data as Objects within Buckets. This flat structure allows it to scale virtually infinitely.



1. Global Infrastructure

S3 is a globally distributed service

Data is stored across:

Regions (e.g., Asia Pacific, US East)

Availability Zones (AZs) within a region

Ensures high availability and fault tolerance

2. Regions & Availability Zones

Each bucket is created in a specific region

Data is automatically replicated across multiple AZs

Example:

Bucket in Mumbai region → data stored across multiple data centers in that region

3. Data Storage Model (Object-Based Storage)

Unlike traditional file systems:

No folders/directories physically

Uses:

Buckets → Containers

Objects → Files

Keys → Unique identifiers

4. S3 Storage Classes

S3 offers a range of storage classes designed for different access patterns and costs.

Storage Class	Access Frequency	Availability Zones	Use Case
S3 Standard	Frequent	≥ 3	General-purpose storage, static websites, cloud apps.
S3 Intelligent-Tiering	Unknown/Changing	≥ 3	Data with unpredictable access patterns. Automatically moves data between tiers to save costs.
S3 Standard-IA	Infrequent	≥ 3	Long-lived data accessed less than once a month (e.g., backups, DR).
S3 One Zone-IA	Infrequent	1	Non-critical, reproducible data (e.g., secondary backups). Cheaper but less durable.
S3 Glacier Instant	Rare (Quarterly)	≥ 3	Archives that need millisecond access (e.g., medical records).
S3 Glacier Flexible	Rare (Yearly)	≥ 3	Archives where retrieval time of minutes/hours is acceptable.
S3 Glacier Deep Archive	Very Rare	≥ 3	Long-term retention (7-10 years) for compliance. Lowest cost.

Buckets

A [bucket](#) is the fundamental container for data in S3. It defines the namespace and controls for the objects it contains.

- **Global Uniqueness:** Bucket names must be **globally unique** across all AWS accounts in a partition (e.g., aws-standard).
- **Regional Ownership:** You choose a specific AWS Region when creating a bucket; data never leaves that region unless explicitly replicated.

Types of Buckets:

1. **General Purpose:** The standard type for most use cases.
2. **Directory Buckets:** Optimized for ultra-low latency (single-digit milliseconds) and high-performance workloads, supporting the **S3 Express One Zone** storage class.
3. **Table/Vector Buckets:** Purpose-built for tabular data (Apache Iceberg) or machine learning vector embeddings.

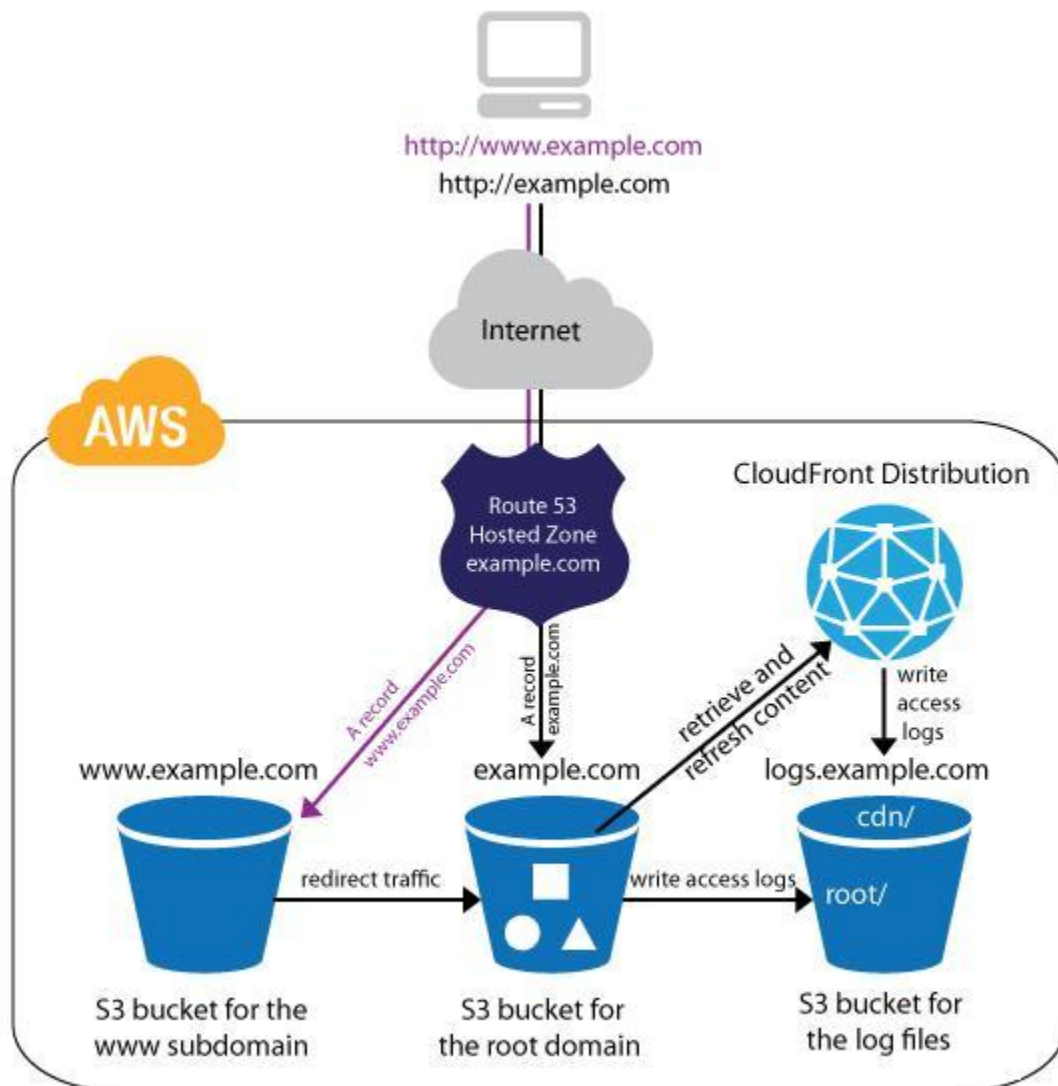
Objects

[Objects](#) are the fundamental entities stored in buckets. An object can range in size from 0 bytes up to **5 TB**.

An object consists of FOUR primary components:

1. **Key:** The unique identifier (name) for the object within the bucket (e.g., images/2024/logo.png). While slashes (/) mimic folders, the structure remains flat—the slash is just part of the string.
2. **Data:** The actual file content (bytes).
3. **Metadata:** A set of name-value pairs describing the object.
 - System Metadata:** Managed by S3 (e.g., Last-Modified, ETag, Size, Storage Class).
 - User-Defined Metadata:** Custom tags or attributes added during upload.
4. **Version ID:** A unique string generated by S3 to distinguish different versions of the same key when **S3 Versioning** is enabled.

The process of hosting a static website using S3 endpoints.



Step 1: Create an S3 Bucket

Go to AWS Console → S3 → Create Bucket

Bucket name must:

Be **globally unique**

Match your domain name (optional but recommended)

step 2: Disable Block Public Access

By default, S3 blocks public access.

Go to:

Bucket → Permissions

Disable:

“Block all public access”

Step 3: Upload Website Files

Upload:

1. index.html (main page)
2. error.html (optional)
3. CSS, JS, images

step 4: Set Bucket Policy (Public Read Access)

Add a policy to allow public access:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "PublicRead",
      "Effect": "Allow",
      "Principal": "*",
      "Action": ["s3:GetObject"],
      "Resource": ["arn:aws:s3:::mywebsite-2026/*"]
    }
  ]
}
```

step 5: Enable Static Website Hosting

Go to:

Properties → Static Website Hosting

Enable it

Enter:

index document: index.html

Error document: error.html

Step 6: Get Website Endpoint URL

After enabling, you get a URL like:

<http://mywebsite-2026.s3-website-ap-south-1.amazonaws.com>

Step 7: Access Your Website

Open the endpoint in a browser

Your static website will load 